

Fluid Motion, Viscosity and Low Pressure

7.1 Introduction

In the case of a liquid at rest, the hydrostatic pressure at any point inside the liquid is given by

$$P = h\rho g$$

However, when the liquid is in a state of motion, to calculate the pressure at a point, certain other factors such as velocity of flow, potential energy, etc., have to be taken into account besides depth and density. A fluid in motion possesses various forms of energy *e.g.* kinetic energy, potential energy and gravitational energy. The viscous property of the fluid and its compressibility have to be taken into account. In the case of ideal liquids, it is assumed that the viscous forces are completely absent and the liquid is highly incompressible.

7.2 Stream Line Motion and Rate of Flow

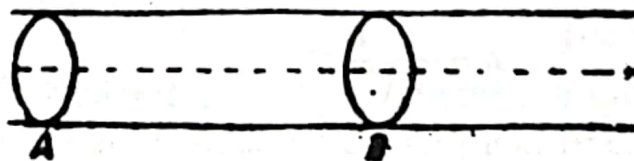


Fig. 7.1

Consider a liquid flowing through a pipe line. Let v be the velocity of flow and a the area of cross-section of the pipe (Fig. 7.1)
Volume of liquid flowing per unit time

$$\begin{aligned} &= \frac{l \times a}{t} = \frac{v \times t \times a}{t} = v \times a \\ &= a \times v \\ &= \text{area of cross section} \times \text{velocity of flow} \end{aligned}$$

The flow of liquid through a pipe line is said to be steady or stream line if at any given point, the velocity of the particle is constant. It means that all the particles of the liquid passing across the point have the same velocity. In the case of stream line motion, at any instant, the tangent to the curve gives the direction of the flow of the liquid. The stream line may be straight or curved. The condition for stream line motion depends on the velocity of flow of the liquid. If the velocity does not exceed a certain limiting value called the critical velocity, flow is said to be stream line. If the velocity of flow is higher than the critical velocity the motion is said to be *turbulent*. For laminar flow, Reynold's number R should be less than 2000. According to Reynold, the critical velocity of a liquid is given by

$$v_c = \frac{R\eta}{\rho D}$$

Here R is Reynold's number

η is the coefficient of viscosity of the liquid

ρ is the density and

D is the diameter of the tube

In the case of turbulent flow the path of the particles during their motion is zig-zag. They do not flow in the same layer. In stream line motion, the particles continue to flow in a particular layer.

Example 7.1. Water flows through a horizontal pipe line of varying cross section at the rate of $0.2 \text{ m}^3/\text{s}$. Calculate the velocity of water at a point where the area of cross section of the pipe is 0.02 m^2 .

Ratio of flow = $a v$

$$v = \frac{\text{Rate of flow}}{a}$$

Here, Rate of flow = $0.2 \text{ m}^3/\text{s}$

$a = 0.02 \text{ m}^2$

$$\therefore v = \frac{0.2}{0.02}$$

$$v = 10 \text{ m/s}$$

7.3 Equation of Continuity

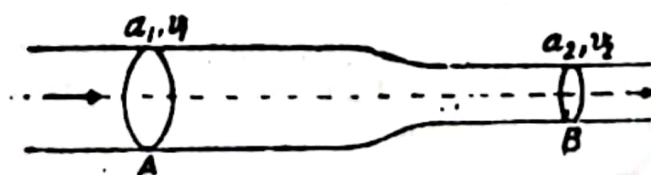


Fig. 7.2

Consider a horizontal pipe (Fig. 7.2). A liquid is flowing through the pipe. The liquid is assumed to be in stream line motion.

Area of cross section at $A = a_1$

Area of cross section at $B = a_2$

As there is no accumulation of the liquid at any point, the amount of liquid flowing per second is the same at all cross-sections of the tube.

Amount of liquid flowing per second at $A = a_1 v_1$

Amount of liquid flowing per second at $B = a_2 v_2$

$$\therefore a_1 v_1 = a_2 v_2 \quad \dots(1)$$

This equation is called the equation of continuity.

In general $a v = \text{constant}$

7.4 Energy of a Liquid in Motion

A liquid in motion possesses three types of energy :

(1) **Kinetic Energy.** The energy is due to the motion of the liquid. It is equal to $\frac{1}{2}mv^2$. The kinetic energy per unit mass of the liquid $= \frac{1}{2}v^2$. The kinetic energy per unit weight is called the *velocity head* and is equal to $\frac{v^2}{2g}$.

(2) **Potential energy.** This energy is due to the position of the liquid with respect to the ground level. It is equal to mgh . The potential energy per unit mass of the liquid $= gh$. The potential energy per unit weight is called the *potential head* and is equal to h .

(3) **Pressure energy.** This energy is due to the pressure exerted on the liquid while it is flowing. The energy is equal to PV . Here P is the pressure and V is the volume of the liquid flowing per unit time.

Pressure energy per unit mass $= \frac{PV}{m} = \frac{P}{\rho}$

Here ρ is the density of the liquid. The pressure energy per unit weight is called the *pressure head* and is equal to $\frac{P}{\rho g}$.

When the flow of the liquid is stream line, the total energy per unit mass of the liquid at any point is given by

$$\left(\frac{v^2}{2} + gh + \frac{P}{\rho} \right)$$

7.5 Bernoulli's Theorem

According to Bernoulli's theorem the sum of the energies possessed by a flowing liquid at any point is constant provided the flow is steady and non turbulent.

Consider an incompressible liquid in a pipe line (Fig. 7.3). The motion of the liquid is stream line. Let the pressures of the liquid at the cross-sections A_1 and A_2 be P_1 and P_2 . The area of cross section at $A_1 = a_1$ and at $A_2 = a_2$. The velocity of flow of the liquid at A_1 is v_1 and at A_2 it is v_2 .

Work done per second on the liquid entering A_1 ,

$$W_1 = P_1 a_1 v_1$$

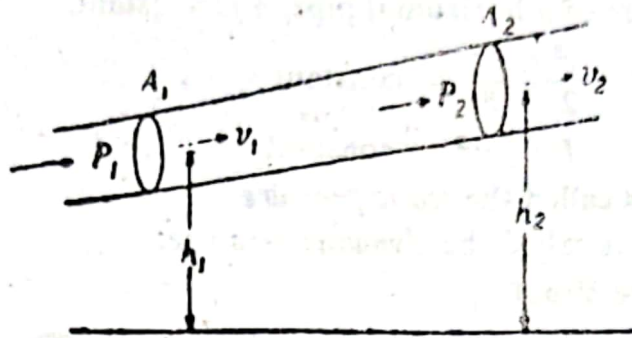


Fig. 7.3

Work done per second by the liquid leaving A_2 ,

$$W_2 = P_2 a_2 v_2$$

$\therefore$ Net work done on the liquid

$$W = W_1 - W_2 = P_1 a_1 v_1 - P_2 a_2 v_2$$

But

$$a_2 v_2 = a_1 v_1$$

$\therefore$

$$W = (P_1 - P_2) a_1 v_1$$

This work done on the liquid contributes for the changes in kinetic energy and gravitational energy.

Change in gravitational energy

$$E_1 = (a_1 v_1) \rho g (h_2 - h_1)$$

Change in kinetic energy

$$E_2 = \frac{1}{2} (a_1 v_1) \rho (v_2^2 - v_1^2)$$

$\therefore$

$$W = E_1 + E_2$$

$\therefore$

$$(P_1 - P_2) a_1 v_1 = (a_1 v_1) \rho g (h_2 - h_1)$$

$$+ \frac{1}{2} (a_1 v_1) \rho (v_2^2 - v_1^2)$$

$$P_1 + \rho g h_1 + \frac{1}{2} \rho v_1^2 = P_2 + \rho g h_2 + \frac{1}{2} \rho v_2^2$$

$$\text{or} \quad \frac{v_1^2}{2} + g h_1 + \frac{P_1}{\rho} = \frac{v_2^2}{2} + g h_2 + \frac{P_2}{\rho} \quad \dots(i)$$

$$\text{or} \quad \frac{v^2}{2} + g h + \frac{P}{\rho} = \text{constant} \quad \dots(ii)$$

This equation represents *Bernoulli's equation*.

Dividing equation (ii) by g

$$\left(\frac{v^2}{2g} \right) + h + \left(\frac{P}{g\rho} \right) = \text{const.} \quad \dots(iii)$$

Here $\frac{v^2}{2g}$ represents the *velocity head*

h represents *gravitational head*

and $\frac{P}{\rho g}$ represents *pressure head*

Special case

In the case of a horizontal pipe, h is constant,

$$\therefore \frac{v^2}{2} + \frac{P}{\rho} = \text{constant}$$

or $P + \frac{1}{2}\rho v^2 = \text{constant}$

Here P is called the *static pressure*

and $\frac{1}{2}\rho v^2$ is called the *dynamic pressure*.

7.6 Alternative Proof

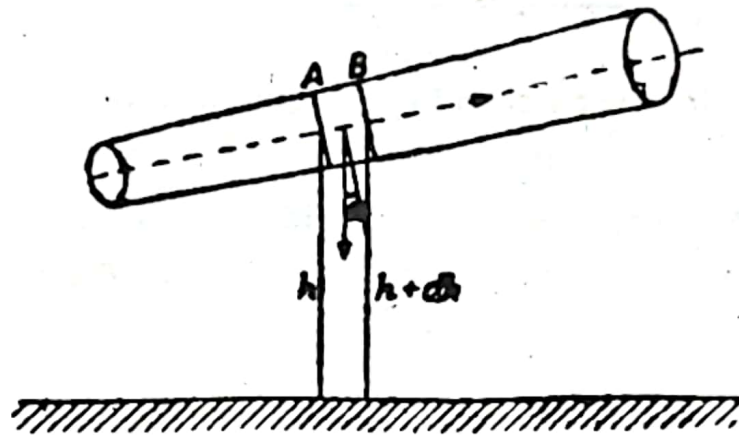


Fig. 7.4

Consider two cross-sections A and B close to each other separated by a distance dx (Fig. 7.4). Assuming the cross-section for the element AB to be uniform.

Volume of the element $= a(dx)$

Mass of the liquid in the element $= \rho(ax)$

Forward force at A along the axis $= Pa$

Backward force at $B = (P + dP)a$

Apart from these two forces, the force acting downward along the axis due to the weight of the element $= \rho(ax)g \sin \theta$.

Here $\sin \theta = \frac{dh}{dx}$

$\therefore$ Net force acting on the liquid in the direction of flow

$$dF = Pa - (P + dP)a - (\rho ax)g \times \frac{dh}{dx}$$

If v is the velocity of flow of the liquid,

$$dF = (m) \frac{dv}{dt} = \rho(ax) \frac{dv}{dt}$$

$$\therefore \rho(ax) \frac{dv}{dt} = -(dP)a - (\rho ax)g \frac{dh}{dx}$$

Arranging and dividing by $a\rho$, we get.

$$\frac{dP}{\rho} + gdh + \left(\frac{dv}{dt}\right) dx = 0$$

Put $\frac{dv}{dt} dx = \left(\frac{dx}{dt}\right) dv = v(dv)$

$$\therefore \frac{dP}{\rho} + gdh + vdv = 0$$

Integrating

$$\frac{P}{\rho} + gh + \frac{v^2}{2} = \text{Constant}$$

or $\frac{v^2}{2} + gh + \frac{P}{\rho} = \text{Constant}$

This represents Bernoulli's equation.

Example 7.2. A Pitot tube is fixed on the wing of an aeroplane to measure the speed of the aeroplane. The tube contains a liquid of density 800 kg/m^3 . The difference in level between the two limbs is 0.5 m . Density of air $= 1.293 \text{ kg/m}^3$. Calculate the speed of the aeroplane.

Here $\frac{1}{2}\rho v^2 = hdg$

$$\rho = 1.293 \text{ kg/m}^3$$

$$d = 800 \text{ kg/m}^3$$

$$h = 0.5 \text{ m}$$

$$g = 9.8 \text{ m/s}^2$$

$$\therefore v^2 = \frac{2h dg}{\rho}$$

$$v = \sqrt{\frac{2hdg}{\rho}} = \sqrt{\frac{2 \times 0.5 \times 800 \times 9.8}{1.293}}$$

$$= 77.85 \text{ m/s.}$$

7.7 Practical Applications

1. Venturimeter

A venturimeter consists of a horizontal tube as shown in Fig. 7.5 It has different cross-sections at A and B . It is used to find the rate of flow of a liquid when the motion of the liquid is steady and non-turbulent. The rate of flow of the liquid crossing through any cross-section of the pipe is constant. It means that the amount of liquid crossing per second at A is equal to the amount of liquid crossing per second at B . But the area of cross section at A is large

and at B it is small. Therefore the velocity of flow of the liquid at A is less and at B it is more.

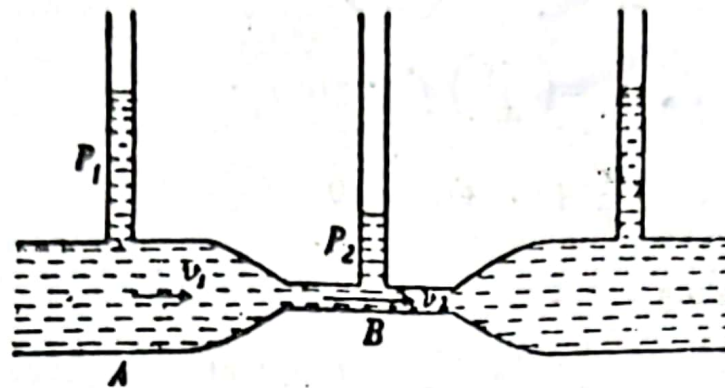


Fig. 7.5

According to Bernoulli's theorem,

$$\frac{v_1^2}{2} + \frac{P_1}{\rho} = \frac{v_2^2}{2} + \frac{P_2}{\rho}$$

Here $v_2 > v_1$; therefore $P_1 > P_2$

The pressure difference at A and B is indicated by the manometer limbs.

Example 7.3. A horizontal tube has different areas of cross section at two points P_1 and P_2 . The diameter at P_1 is 0.04 m and the diameter at P_2 is 0.02 m. Two manometer limbs are fixed at P and Q . When a liquid of density 800 kg/m³ flows through, the difference in pressure between the manometer limbs is 0.08 m. Calculate the rate of flow of the liquid through the tube.

According to Bernoulli's principle,

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho gh_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho gh_2$$

Here $h_1 = h_2$

$$\therefore P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$$

or $(P_1 - P_2) = \frac{1}{2}\rho(v_2^2 - v_1^2)$

But $a_1 v_1 = a_2 v_2$

or $v_2 = \frac{a_1 v_1}{a_2}$

$$\therefore (P_1 - P_2) = \frac{1}{2}\rho \left[\left(\frac{a_1 v_1}{a_2} \right)^2 - v_1^2 \right]$$

or $v_1^2 = \frac{2(P_1 - P_2)a_2^2}{\rho(a_1^2 - a_2^2)}$

or $v_1 = a_2 \sqrt{\frac{2(P_1 - P_2)}{\rho(a_1^2 - a_2^2)}}$

But rate of flow $= a_1 v_1$

$$= a_1 a_2 \sqrt{\frac{2(P_1 - P_2)}{\rho(a_1^2 - a_2^2)}}$$

Here $(P_1 - P_2) = 0.08 \text{ m of the liquid column}$

$$= 0.08 \times 800 \times 9.8 \text{ N/m}^2$$

$$a_1 = \pi(0.02)^2 = 4\pi \times 10^{-4} \text{ m}^2$$

$$a_2 = \pi(0.01)^2 = \pi \times 10^{-4} \text{ m}^2$$

$$\rho = 800 \text{ kg/m}^3.$$

$\therefore$ Rate of flow

$$= 4\pi \times 10^{-4} \times \pi \times 10^{-4} \sqrt{\frac{2 \times 0.08 \times 800 \times 9.80}{800(16\pi^2 \times 10^{-8} - \pi^2 \times 10^{-8})}}$$

$$= 4.062 \times 10^{-4} \text{ m}^3/\text{s}$$

2. Bunsen Burner

In a bunsen burner, the gas enters the base and comes out of the nozzle N . As the cross-section of the nozzle is very small, the velocity of the gas increases. Consequently, pressure just above the nozzle N is decreased (Fig. 7.6). Air from the atmosphere rushes into the burner. This mixture of gas and air moves up the burner and burns at the top.

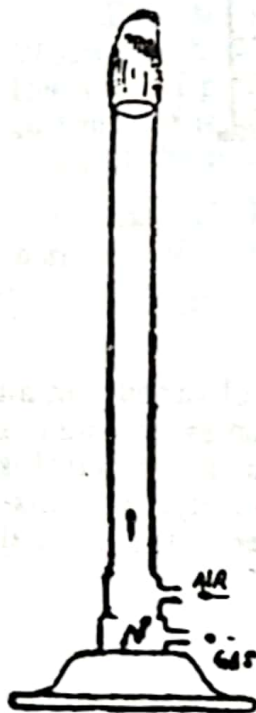


Fig. 7.6

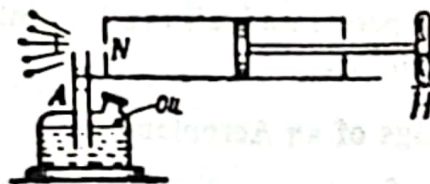


Fig. 7.7

3. Atomiser or Sprayer

An atomizer is shown in Fig. 7.7. The oil is kept in a can and it is fitted with a vertical tube A . When the handle of the pump pushes the piston, air comes out through the nozzle N with a high velocity. As the velocity increases, the pressure just outside N decreases and the oil in the tube A is pushed up due to the atmospheric pressure acting on the surface of the liquid in the can. When the oil reaches the top of the tube A , it is sprayed out by the escaping air through N .

$$\begin{aligned}\text{Rate of flow} &= (0.09\pi \times 0.01\pi) \sqrt{\frac{2 \times 9.8 \times 10^3}{10^3 \times 8 \times 10^{-3} \pi^2}} \\ &= 0.1397 \text{ m}^3/\text{s}\end{aligned}$$

Example 7.10 (A). Air is streaming past a horizontal airplane wing such that its speed is 120 ms^{-1} over the upper surface and 90 ms^{-1} at the lower surface. If the density of air is 1.3 kg m^{-3} , find the difference in pressure between the top and the bottom of the wing. If the wing is 10 m long and has an average width of 2 m , calculate the gross lift of the wing. [Roorkee Engineering Entrance Exam., 1983]

$$\frac{p_1}{\rho} + gh_1 + \frac{1}{2} v_1^2 = \frac{p_2}{\rho} + gh_2 + \frac{1}{2} v_2^2$$

$$\text{As } h_1 = h_2$$

$$\frac{p_1}{\rho} + \frac{1}{2} v_1^2 = \frac{p_2}{\rho} + \frac{1}{2} v_2^2$$

$$\text{or } (p_1 - p_2) = \frac{1}{2} \rho (v_2^2 - v_1^2)$$

Gross lift of the wing

$$L = (p_1 - p_2) A = \frac{1}{2} \rho (v_2^2 - v_1^2) A$$

$$\text{Here } v_2 = 120 \text{ m/s}$$

$$v_1 = 90 \text{ m/s}$$

$$\rho = 1.3 \text{ kg/m}^3$$

$$\therefore \text{Lift} = \frac{1}{2} \times 1.3 \times [(120)^2 - (90)^2] \times 20$$

$$\text{Lift} = 81900 \text{ N}$$

7.8 Torricelli's Theorem

According to Torricelli's theorem, the velocity of efflux of a liquid through an orifice is equal to the velocity which a freely falling body will acquire while falling through a height equal to the height of the free surface of the liquid above the orifice (Fig. 7.11).

According to Bernoulli's theorem

$$\frac{v^2}{2} + gh + \frac{P}{\rho} = \text{Constant}$$

As P is the same at the free surface and at the orifice,

$$\frac{v^2}{2} + gh = \text{Constant}$$

$$\text{At A, } v = 0, \quad h = h_1$$

$$\text{At B, } h = 0$$

$$\therefore gh_1 = \frac{v^2}{2} \quad \text{or} \quad v = \sqrt{2gh_1} \quad \dots(i)$$

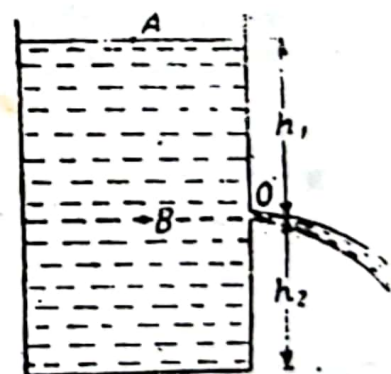


Fig. 7.11

The horizontal range,

$$l = v \times t$$

$$l = v \times \sqrt{\frac{2h_2}{g}}$$

$$l = \left(\sqrt{2gh_1} \right) \sqrt{\frac{2h_2}{g}}$$

$$l = 2\sqrt{h_1 h_2} \quad \dots(ii)$$

But

$$h_2 = H - h_1$$

$$l = 2\sqrt{h_1 (H - h_1)}$$

$$l^2 = 4h_1 H - 4h_1^2$$

Differentiating

$$2l \left(\frac{dl}{dh_1} \right) = 4H - 8h_1$$

For l to be maximum, $\frac{dl}{dh_1} = 0$

$$\therefore 4H - 8h_1 = 0$$

$$h_1 = \frac{H}{2}$$

and

$$h_2 = \frac{H}{2}$$

Therefore, for the range of the liquid to be maximum, the orifice must be at half the height of the liquid column.

Example 7.11. A tank containing water has an orifice on one vertical side. If the centre of the orifice is 4.9 metres below the surface level in the tank, find the velocity of discharge, assuming that there is no wastage of energy. (Delhi 1976)

$$\frac{1}{2}mv^2 = mgh$$

$$v = \sqrt{2gh}$$

Here $g = 9.8 \text{ m/s}^2$ and $h = 4.9 \text{ m}$

$$\therefore v = \sqrt{2 \times 9.8 \times 4.9}$$

$$v = 9.8 \text{ m/s}$$

7.9 Viscosity

Whenever a liquid flows on a horizontal surface, the velocities of the different layers of the liquid parallel to the fixed surface are different and increase with the distance from the fixed surface. If the motion of the liquid is streamline, the layer of the liquid in contact with the fixed surface is stationary. The velocity of any layer increases with the distance from the fixed layer. If any two layers are considered, the upper layer tends to accelerate the motion of the lower layer and the lower

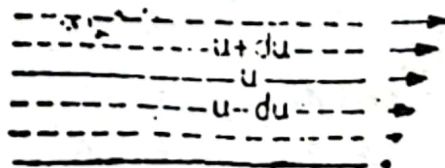


Fig. 7.12

layer tends to retard the motion of the upper layer. The two layers together tend to destroy their relative motion, as if there is a backward dragging tangential force (Fig. 7.12). An external force is required to overcome this backward drag and to maintain the relative velocity between the different layers of the liquid. This property by virtue of which a liquid opposes the relative motion between the different layers is called viscosity or internal friction. Generally, thin liquids like alcohol, water, spirit etc., are less viscous whereas thick liquids like coal tar, castor oil, glycerine etc., are more viscous.

According to Newton, the backward tangential force F on any layer is dependent on

(i) $F \propto A$, the area of the layer

(ii) $F \propto \frac{du}{dx}$ the velocity gradient at the layer

$$F \propto A \frac{du}{dx}$$

or $F = -\eta A \frac{du}{dx}$

where η is called the coefficient of viscosity of the fluid and $\frac{du}{dx}$ is called the velocity gradient or change of velocity with distance. The negative sign shows that the force is acting opposite to the direction of velocity.

If $A = 1$ and $\frac{du}{dx} = 1$

then, $F = -\eta$

Therefore, the coefficient of viscosity is defined as the tangential force required to maintain a unit velocity gradient between two layers of area one unit each.

Units. If $A = 1 \text{ sq cm}$, $\frac{du}{dx} = \frac{1 \text{ cm/s}}{1 \text{ cm}}$

and $F = 1 \text{ dyne}$, then

$$\eta = \frac{F}{A \cdot \frac{du}{dx}}$$

Numerically $\eta = 1 \text{ poise}$

Poise. The coefficient of viscosity of a liquid is one poise if a force of 1 dyne is required to maintain a velocity gradient of one unit between two layers of area 1 sq cm each.

Dimensions of coefficient of viscosity

$$[\eta] = \frac{[F]}{[A] \left[\frac{du}{dx} \right]} = \frac{[MLT^{-2}]}{[L^2][T^{-1}]} = [ML^{-1}T^{-1}]$$

Unit of η in SI is kg/m-s or N-s/m²

7.10 Stoke's Law

When a small steel ball is left gently on the surface of a long vertical column of a viscous liquid, the ball moves vertically downward. Initially, the ball is accelerated due to gravity. The motion of the ball is opposed by the viscosity of the liquid. Consequently, the resultant force acting on the ball becomes zero and it moves with a constant velocity called the *terminal velocity* v .

The resistive force F , acting on the ball depends on

- (i) the radius of the ball, r
- (ii) the terminal velocity, v and
- (iii) coefficient of viscosity, η

Expressing dimensionally

$$[F] = k[r^a][v^b][\eta^c] \quad \dots(i)$$

$$[MLT^{-2}] = [L]^a [LT^{-1}]^b [ML^{-1}T^{-1}]^c$$

$$[MLT^{-2}] = [M]^c [L]^{a+b-c} [T]^{-b-c}$$

Equating the powers

$$c = 1$$

$$-b-c = -2$$

$$\therefore b = 1$$

$$a+b-c = 1$$

$$a = 1$$

Substituting these values in equation (i)

$$F = k r v \eta \quad \dots(ii)$$

But $k = 6\pi$

$$\therefore F = 6\pi \eta r v \quad \dots(iii)$$

Coefficient of viscosity of a liquid can be determined by Stoke's formula.

Suppose,

Radius of the ball $= r$

Terminal velocity $= v$

Coefficient of viscosity $= \eta$

Density of the ball $= \rho$

Density of the liquid $= \sigma$

$$\text{Net weight of the ball} = \frac{4}{3} \pi r^3 (\rho - \sigma)g \quad \dots(ii)$$

$$\text{Upward force due to viscosity} = 6\pi\eta r v \quad \dots(iii)$$

$$\therefore 6\pi\eta r v = \frac{4}{3} \pi r^3 (\rho - \sigma) g$$

$$v = \frac{2r^2 (\rho - \sigma) g}{9\eta} \quad \dots(iv)$$

$$\eta = \frac{2r^2 (\rho - \sigma) g}{9v} \quad \dots(v)$$

Hence η can be determined.

Stoke's law is applicable to determine the coefficient of viscosity of highly viscous liquids.

Example 7.11 (a). Find the limiting velocity of a rain drop.

Assume, diameter = 10^{-3} m

Density of air relative to water = 1.3×10^{-3}

Coefficient of viscosity of air = 1.81×10^{-6} S.I. units

Density of water 10^3 kg/m³

[Rajasthan, 1981]

According to Stoke's law

$$6\pi\eta r v = \frac{4}{3} \pi r^3 (\rho - \sigma) g$$

$$v = \frac{2r^2 (\rho - \sigma) g}{9\eta}$$

$$v = \frac{2r^2 \rho \left[1 - \left(\frac{\sigma}{\rho} \right) \right] g}{9\eta}$$

Here

$$r = 5 \times 10^{-4} \text{ m}$$

$$\rho = 10^3 \text{ kg/m}^3$$

$$\frac{\sigma}{\rho} = 1.3 \times 10^{-3}$$

$$\eta = 1.81 \times 10^{-6} \text{ S. I. units}$$

$$v = \frac{2 \times (5 \times 10^{-4})^2 \times 10^3 [1 - 1.3 \times 10^{-3}] \times 9.8}{9 \times 1.81 \times 10^{-6}}$$

$$v = 30.04 \text{ m/s.}$$

Example 7.12. Two equal drops of water (surface tension T), each of radius r , are falling through air (viscosity η) with a steady velocity v . If the two drops coalesce to form a bigger drop, (i) Compute the energy released and (ii) find the new velocity of fall.

[Delhi, Engineering 1980]

Let the radius of the bigger drop be R

$$\therefore \frac{4}{3} \pi R^3 = 2 \left(\frac{4}{3} \pi r^3 \right)$$

$$R = (2)^{1/3} r = 1.26 r$$

(i) Decrease in surface area

$$= 2(4\pi r^2) - 4\pi R^2$$

$$\begin{aligned}
 &= 4\pi [2r^2 - R^2] = 4\pi [2r^2 - (1.26r)^2] \\
 &= 4\pi [2r^2 - 1.588r^2] = 4\pi \times 0.412r^2 \\
 &= 5.178r^2
 \end{aligned}$$

$$\text{Energy released} = T \times 5.178r^2 = 5.178 Tr^2$$

(2) Let the velocity of the bigger drop be V

For the small drop,

$$mg = 6\pi\eta rv \quad \dots(i)$$

For the bigger drop

$$2mg = 6\pi\eta RV \quad \dots(ii)$$

Dividing (ii) by (i)

$$2 = \frac{RV}{rv}$$

or

$$V = \frac{2rv}{R}$$

But

$$R = 1.26r$$

$\therefore$

$$V = \frac{2rv}{1.26r} = 1.587 v \quad \dots(iii)$$

7.11. Poiseuille's Method for Coefficient of Viscosity

Rate of flow of a liquid through a capillary tube depends upon

(a) length of the capillary tube

(b) radius of the capillary tube

(c) pressure difference between two ends of the capillary tube

and

(d) Coefficient of viscosity of the liquid.

It is also assumed that (i) the flow of the liquid is stream line (ii) the liquid in contact with the walls of the tube is at rest (iii) the tube is kept horizontal and (iv) the pressure at any given cross section is constant and there is no radial flow of the liquid.

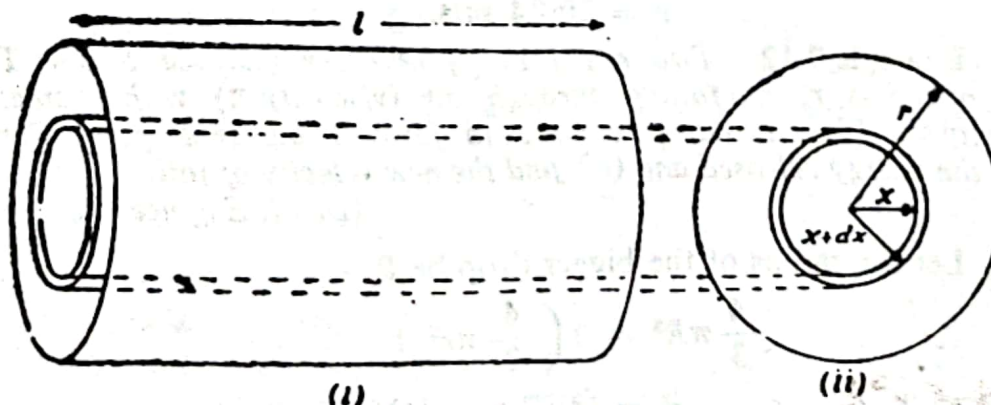


Fig. 7.13

Consider a liquid flowing in a capillary tube AB of

length l and radius r (Fig. 7.13). The layer at the centre of the tube has maximum velocity and the velocity decreases from the axis to the walls of the tube. Let a layer be at a distance x from the axis. The other layer is at a distance $x+dx$ from the axis. The tangential force acting in the opposite direction to the direction of flow is given by

$$F = -\eta A \frac{du}{dx}$$

Here

$$A = 2\pi x l$$

$$\therefore F = \eta (2\pi x) l \frac{du}{dx} \quad \dots(i)$$

The force at the end A of the tube,

$$F = P (\pi x^2) \quad \dots(ii)$$

where P is the difference in pressure between the two ends of the capillary tube.

For steady flow, equating (i) and (ii)

$$-\eta (2\pi x l) \frac{du}{dx} = P \pi x^2$$

$$\text{or} \quad \frac{du}{dx} = -\frac{Px}{2\eta l}$$

$$\text{or} \quad du = -\left(\frac{P}{2\eta l}\right) x dx \quad \dots(iii)$$

Integrating,

$$du = -\frac{Px^2}{4\eta l} + k$$

$$\text{When } x = r, \quad u = 0$$

$$\therefore k = \frac{Pr^2}{4\eta l}$$

$$\therefore u = -\frac{Px^2}{4\eta l} + \frac{Pr^2}{4\eta l} = \frac{P(r^2 - x^2)}{4\eta l} \quad \dots(iv)$$

Equation (iv) gives the velocity of the liquid for a layer of radius x .

Volume of the liquid between the two layers per second
 $= dV$

$$dV = (2\pi x) dx \cdot u$$

$$dV = \frac{(2\pi x dx) P (r^2 - x^2)}{4\eta l}$$

Rate of flow of the liquid through the whole tube is given by

$$\int dV = \int_0^r \frac{(2\pi x dx) P (r^2 - x^2)}{4\eta l}$$

$$V = \frac{\pi P}{2 \eta l} \int_0^r (r^2 x - x^3) dx$$

$$V = \frac{\pi P}{2 \eta l} \left[\frac{r^4}{2} - \frac{r^4}{4} \right] = \frac{\pi P r^4}{8 \eta l} \quad \dots(v)$$

or

$$\eta = \frac{\pi P r^4}{8 l V} \quad \dots(vi)$$

In the experiment, a capillary tube AB is fixed as shown in Fig. 7-14. The amount of liquid flowing per unit time is measured. Let it be V

Here $P = h \rho g$

The radius of the tube is measured with the help of a travelling microscope. The difference in the levels of the manometer gives h .

From equation, (vi), η can be calculated.

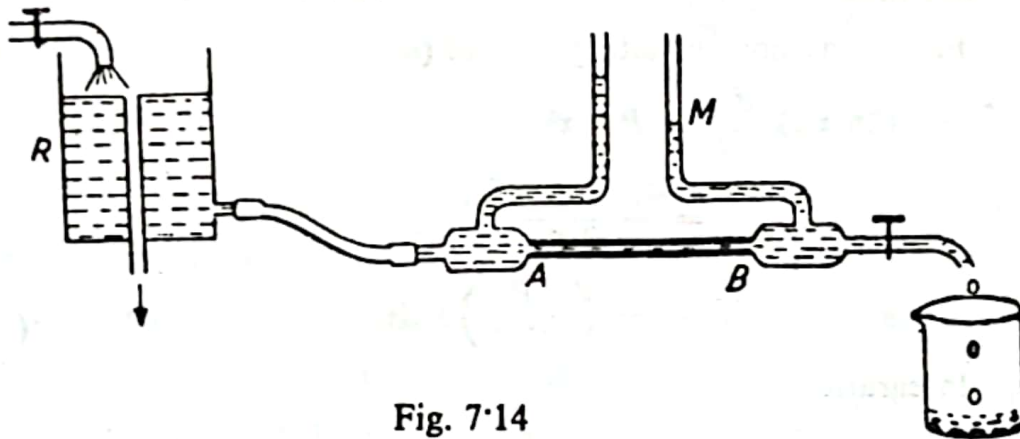


Fig. 7-14

7-12 Correction to Poiseuille's Equation

In the Poiseuille's equation two important corrections are to be applied.

(i) In deriving the equation, it has been assumed that the liquid does not possess any kinetic energy at the outlet end of the tube. However the liquid flowing out of the tube has a certain velocity and hence possesses kinetic energy. The effective pressure P_1 is given by

$$P_1 = P - P' \quad \dots(i)$$

Kinetic energy given to the liquid per second

$$E' = \int_0^r \frac{1}{2} m u^2 = \int_0^r \frac{1}{2} \left[(2\pi x dx) u \rho \right] u^2$$

$$E' = \pi \rho \int_0^r u^3 x dx$$

But $u = \frac{P(r^2 - x^2)}{4 \eta l}$

$$\begin{aligned}
 \therefore E' &= \pi \rho \int_0^r \left(\frac{P}{4\eta l} \right)^3 (r^2 - x^2)^3 x dx \\
 E' &= \pi \rho \left(\frac{P}{4\eta l} \right)^3 \int_0^r (r^6 x - 3r^4 x^3 + 3r^2 x^5 - x^7) dx \\
 E' &= \pi \rho \left[\frac{P}{4\eta l} \right]^3 \left(\frac{r^8}{8} \right) = \left(\frac{\pi P r^4}{8\eta l} \right)^3 \frac{\rho}{\pi^2 r^4} \\
 F' &= \frac{V^3 \rho}{\pi^2 r^4} \\
 \therefore P' V &= E' \\
 P' &= \frac{E'}{V} = \frac{V^2 \rho}{\pi^2 r^4} \\
 \therefore P_1 &= P - \left(\frac{V^2 \rho}{\pi^2 r^4} \right) \quad \dots(ii)
 \end{aligned}$$

If V is negligibly small, $P_1 = P$

(ii) At the inlet end of the tube, the flow of the liquid is not stream line for some distance. Consequently the liquid is accelerated. The effective length of the tube is given by $(l + 1.64 r)$.

$$\begin{aligned}
 \therefore \eta &= \frac{\pi P r^4}{8V(l + 1.64r)} - \left(\frac{V^2 \rho}{\pi^2 r^4} \right) \frac{\pi r^4}{8V(l + 1.64r)} \\
 \eta &= \frac{\pi P r^4}{8V(l + 1.64r)} - \frac{V\rho}{8\pi(l + 1.64r)} \quad \dots(iii)
 \end{aligned}$$

Example 7.13 Water flows through a horizontal capillary tube of 1 mm internal diameter and length 70 cm under pressure of a column of water 30 cm in height. Find the rate of flow of water through the capillary tube. Viscosity of water = 10^{-3} N-s/m². [Calcutta 1978]

Rate of flow,

$$\begin{aligned}
 V &= \frac{\pi P r^4}{8\eta l} \\
 \text{Here } P &= 0.3 \text{ m of water} = 0.3 \times 10^3 \times 9.8 \text{ N/m}^2 \\
 r &= 0.5 \text{ mm} = 5 \times 10^{-4} \text{ m} \\
 l &= 0.7 \text{ m} \\
 \eta &= 10^{-3} \text{ N-s/m}^2 \\
 \therefore V &= \frac{\pi \times 0.3 \times 10^3 \times 9.8 \times [5 \times 10^{-4}]^4}{1 \times 10^{-3} \times 0.7} \\
 V &= 1.0312 \times 10^{-7} \text{ m}^3/\text{s}
 \end{aligned}$$

7.13 Ostwald's Viscometer

Ostwald's viscometer is used to compare the coefficients of viscosity of two different liquids. It is also used to compare the coefficients of viscosity of the same liquid at different temperatures

$$h_2 = \frac{h_1}{2} \quad \text{or} \quad \frac{h_1}{h_2} = 2$$

$$T = ?$$

$$T = \left[\frac{8 \times 25 \times 10^{-4} \times \pi \times 10^{-3} \times 0.2 \times 2.3026}{\pi \times 10^3 \times 9.80 \times (5 \times 10^{-4})^4} \right] \log(2)$$

$$T = 4526 \text{ s} = 1.257 \text{ hours.}$$

Example. 7.16. A wide vertical tube has a short capillary tube fixed horizontally at its lower end. When it is filled with glycerine it takes 45 seconds for the level to fall between two fixed marks. When the tube is filled with oil it takes 67.5 seconds for the level to fall between the same two fixed marks. Calculate the coefficient of viscosity of oil. Given, density of glycerine = $1.2 \times 10^3 \text{ kg/m}^3$, density of oil = $0.8 \times 10^3 \text{ kg/m}^3$, viscosity of glycerine = 0.85 N-s/m^2 .

$$V = \frac{\pi P_1 r^4}{8 \eta_1 l}$$

$$V_2 = \frac{\pi P_2 r^4}{8 \eta_2 l}$$

$$V_1 t_1 = V_2 t_2$$

$$\left(\frac{\pi P_1 r^4}{8 \eta_1 l} \right) t_1 = \left(\frac{\pi P_2 r^4}{8 \eta_2 l} \right) t_2$$

$$\frac{\eta_2}{\eta_1} = \left(\frac{P_2}{P_1} \right) \left(\frac{t_2}{t_1} \right) = \left(\frac{h P_2 g}{h P_1 g} \right) \left(\frac{t_2}{t_1} \right)$$

$$\eta_2 = \eta_1 \left(\frac{\rho_2}{\rho_1} \right) \left(\frac{t_2}{t_1} \right)$$

Here,

$$\eta_1 = 0.85 \text{ N-s/m}^2, \rho_2 = 0.8 \times 10^3 \text{ kg/m}^3$$

$$\rho_1 = 1.2 \times 10^3 \text{ kg/m}^3, t_2 = 67.5 \text{ s,}$$

$$t_1 = 45 \text{ s}$$

$\therefore$

$$\eta_2 = 0.85 \left(\frac{0.8 \times 10^3}{1.2 \times 10^3} \right) \left(\frac{67.5}{45} \right)$$

$$\eta_2 = 0.85 \text{ N-s/m}^2$$

Example 7.17. Water is conveyed through a horizontal tube 0.08 m in diameter and 4 km in length at the rate of 20 litres per second. Assuming only viscous resistance, calculate the pressure difference required to maintain the flow ($\eta = 10^{-3} \text{ N-s/m}^2$)

(Delhit 1976)

According to Poiseuille's formula

$$V = \frac{\pi P r^4}{8 \eta l}$$

$$P = \frac{8 V \eta l}{\pi r^4}$$

$$V = 20 \text{ litres/s} = 20 \times 10^{-3} \text{ m}^3/\text{s}$$

$$\eta = 10^{-3} \text{ N-s/m}^2$$

$$l = 4 \text{ km} = 4 \times 10^3 \text{ m}$$

$$r = 0.04 \text{ m}$$

$$P = \frac{8 \times 20 \times 10^{-3} \times 10^{-8} \times 4 \times 10^3}{\pi (0.04)^4}$$

$$P = 1.7956 \times 10^4 \text{ N/m}^2$$

Example. 7.18. A plate of metal 10^{-2} m^2 area rests on a layer of castor oil $2 \times 10^{-3} \text{ m}$ thick, whose coefficient of viscosity is 1.55 N-s/m^2 . Calculate the horizontal force required to move the plate with a uniform speed of $3 \times 10^{-2} \text{ m/s}$.

(Gorakhpur, 1971, 1974, Delhi, 1977)

$$F = \eta A \frac{du}{dx}$$

Here

$$\eta = 1.55 \text{ N-s/m}^2$$

$$A = 10^{-2} \text{ m}^2$$

$$\frac{du}{dx} = \frac{3 \times 10^{-2}}{2 \times 10^{-3}} = 15 \text{ per second}$$

$$F = 1.55 \times 10^{-2} \times 15$$

$$F = 0.2325 \text{ N}$$

Example 7.19. A large bottle is fitted with a siphon made of capillary glass tubing. Compare the coefficients of viscosity of water and petrol if the time taken to empty the bottle in the two cases is in the ratio 2 : 5. Specific gravity of petrol = 0.8. (Delhi, 1978)

$$V_1 = \frac{\pi P_1 r^4}{8 \eta_1 l}$$

$$V_2 = \frac{\pi P_2 r^4}{8 \eta_2 l}$$

$$V_1 = \frac{V}{t_1}$$

$$V_2 = \frac{V}{t_2}$$

$$\frac{V_1}{V_2} = \frac{t_2}{t_1} = \frac{5}{2}$$

Also

$$\frac{V_1}{V_2} = \frac{P_1 \eta_2}{P_2 \eta_1}$$

Here

$$\frac{P_1}{P_2} = \frac{d_1}{d_2} = \frac{10^3}{0.8 \times 10^3} = 1.25$$

$\therefore$

$$\frac{5}{2} = 1.25 \times \frac{\eta_2}{\eta_1}$$

$$\frac{\eta_1}{\eta_2} = \frac{1.25 \times 2}{5} = 0.5$$

$$\eta_1 : \eta_2 \therefore 1 : 2$$

Example. 7.20. A capillary tube 10^{-3} m in diameter and 0.2 m in length is fitted horizontally to a vessel kept full of alcohol of density $0.8 \times 10^3 \text{ kg/m}^3$. The depth of the centre of the capillary tube below the surface of alcohol is 0.3 m . Viscosity of alcohol is 0.0012 N-s/m^2 .

Calculate the volume of alcohol that flows in 5 minute.

(Gorakhpur, 1977)

$$V = \frac{\pi P r^4}{8 \eta l}$$

Here

$$P = h \rho g = 0.3 \times 0.8 \times 10^3 \times 9.8 \text{ N/m}^2$$

$$r = 5 \times 10^{-4} \text{ m}$$

$$\eta = 0.0012 \text{ N-s/m}^2$$

$$l = 0.2 \text{ m}$$

$$t = 5 \text{ minutes} = 300 \text{ s}$$

$$\begin{aligned} \text{Total volume} &= V \times t = \frac{\pi P r^4 \times t}{8 \eta l} \\ &= \frac{\pi \times 0.3 \times 0.8 \times 10^3 \times 9.8 \times (5 \times 10^{-4})^4 \times 300}{8 \times 0.0012 \times 0.2} \\ &= 7.216 \times 10^{-5} \text{ m}^3 \end{aligned}$$

Example. 7.21. Calculate the mass of water flowing in 10 seconds through a horizontal capillary tube of circular cross section of radius 10^{-3} m . The tube is fitted at the bottom of a constant level tank at a depth of 1 m. Length of the tube is 0.3142 m.

$$\eta = 10^{-3} \text{ N-s/m}$$

(Gorakhpur, 1971)

$$V = \frac{\pi P r^4}{8 \eta l}$$

Mass of water flowing in t seconds

$$M = V \rho t = \frac{\pi P r^4 \rho t}{8 \eta l}$$

Here

$$P = 1 \times 10^3 \times 9.8 \text{ N/m}^2$$

$$r = 10^{-3} \text{ m}$$

$$\rho = 10^3 \text{ kg/m}^3$$

$$t = 10 \text{ s}$$

$$\eta = 10^{-3} \text{ N-s/m}^2$$

$$l = 0.3142 \text{ m}$$

$$M = \frac{3.142 \times 10^3 \times 9.8 \times (10^{-3})^4 \times 10^3 \times 10}{8 \times 10^{-3} \times 0.3142}$$

$$M = 0.1225 \text{ kg}$$

Example 7.22. A spherical glass ball of mass $1.34 \times 10^{-4} \text{ kg}$ and diameter $4.4 \times 10^{-3} \text{ m}$ takes 6.4 s to fall steadily through a height of 0.381 m inside a large volume of oil of specific gravity 0.943. Calculate the coefficient of viscosity of the oil.

(IAS 1978)

$$F = 6\pi \eta r v$$

$$mg - \frac{4}{3} \pi r^3 \rho g = 6\pi \eta r v$$

$$\eta = \frac{Mg - \frac{4}{3} \pi r^3 \rho g}{6\pi r v}$$

Here

$$v = \frac{0.381}{6.4} = 0.05953 \text{ m/s}$$

$$\eta = 1.34 \times 10^{-4} \text{ kg},$$

$$r = 2.2 \times 10^{-3} \text{ m}$$

$$\rho = 0.943 \times 10^3 \text{ kg/m}^3$$

$$g = 9.8 \text{ m/s}^2$$

$$\therefore \eta = \frac{1.34 \times 10^{-4} \times 9.8 - \frac{4}{3} \pi (2.2 \times 10^{-3})^3 \times 0.943 \times 10^3 \times 9.8}{6\pi \times 2.2 \times 10^{-3} (0.05953)}$$

$$\eta = \frac{1.313 \times 10^{-3} - 0.4111 \times 10^{-3}}{2.469 \times 10^{-3}} = \frac{0.9019}{2.469}$$

$$\eta = 0.3653 \text{ N-s/m}^2$$

Example 7.23. Two capillary tubes LM and MN are joined end to end at M. Length of LM is 0.16 m and its diameter is 4×10^{-3} m. Length of MN is 0.04 m and its diameter is 2×10^{-3} m. The composite tube is held horizontally with L connected to a vessel of water having a constant head of 0.3 m. N is open to the atmosphere. Calculate the pressure at M.

(I A S 1976)

Suppose the pressure difference between M and N is $p_2 \text{ N/m}^2$.

Pressure difference between L and M = $(p_1 - p_2) \text{ N/m}^2$

Pressure difference between L and N

$$= p_1 = 0.3 \text{ m of water}$$

$$= 0.3 \times 10^3 \times 9.8 \text{ N/m}^2$$

Rates of flow are equal

$$V_1 = V_2$$

$$V_1 = \frac{\pi (p_1 - p_2) r_1^4}{8 \eta l_1}$$

$$V_2 = \frac{\pi p_2 r_2^4}{8 \eta l_2}$$

But

$$V_1 = V_2$$

$$\frac{\pi (p_1 - p_2) r_1^4}{8 \eta l_1} = \frac{\pi p_2 r_2^4}{8 \eta l_2}$$

$$\frac{p_1 - p_2}{p_2} = \left(\frac{r_2^4 l_1}{r_1^4 l_2} \right)$$

$$\left(\frac{p_1}{p_2} - 1 \right) = \frac{r_2^4 l_1}{r_1^4 l_2}$$

$$\frac{p_1}{p_2} = \frac{r_2^4 l_1 + r_1^4 l_2}{r_1^4 l_2}$$

$$p_2 = \frac{p_1 r_1^4 l_2}{(r_2^4 l_1 + r_1^4 l_2)}$$

$$r_1 = 2 \times 10^{-3} \text{ m}$$

$$r_2 = 10^{-3} \text{ m}$$

$$l_2 = 0.16 \text{ m}$$

$$l_1 = 0.04 \text{ m}$$

$$P_2 = \frac{0.3 \times 10^3 \times 9.8 \times (2 \times 10^{-3})^4 \times 0.04}{(10^{-3})^4 \times 0.16 + (2 \times 10^{-3})^4 \times 0.04}$$

$$P_2 = 2.352 \times 10^3 \text{ N/m}^2$$

7.16. Meyer's Modification of Poiseuille's Formula

Let a gas flow through a capillary tube under steady state condition. Since there is no accumulation of gas at any particular cross-section, the mass of the gas flowing per second across any cross section is constant.

Consider an element of length dx at a distance x from the inlet end of the capillary tube. Let the pressure difference be dP . The volume of the gas flowing per second across the element is given by

$$V = \frac{\pi r^4}{8\eta} \left[-\frac{dP}{dx} \right]$$

or
$$V = -\frac{\pi r^4}{8\eta} \left[\frac{dP}{dx} \right] \quad \dots(i)$$

Here $\frac{dP}{dx}$ represents the pressure gradient along the length of the tube and the negative sign shows that P decreases with increase in x .

Multiplying equation (i) by P on both the sides,

$$PV = -\left(\frac{\pi r^4}{8\eta}\right) \left(P \frac{dP}{dx}\right) = \text{Constant} \quad \dots(ii)$$

PV is constant for all cross-sections of the tube.

Let P_1 be the pressure at the inlet end and P_2 the pressure at the outlet end.

$$(PV) \int_0^l dx = -\left(\frac{\pi r^4}{8\eta}\right) \int_{P_1}^{P_2} P dP$$

$$PVI = -\frac{\pi r^4}{8\eta} \left[\frac{P_2^2 - P_1^2}{2} \right]$$

$$PV = \frac{\pi r^4}{16\eta l} [P_1^2 - P_2^2] \quad \dots(iii)$$

Ar
$$PV = P_1 V_1 = P_2 V_2$$

$$PV = P_1 V_1 = P_2 V_2 = \frac{\pi r^4 [P_1^2 - P_2^2]}{16\eta l} \quad \dots(iv)$$

Equation (iv) represents Meyer's formula for the flow of a gas through a capillary tube.